

# Inferance

2017

81

Relative Efficiency:-

Defination:-

Relative efficiency is define as the ratio of the variance of an estimator or methods. it measure how much more efficient one estimator or method is compared to another

formula:-

$$\text{Relative Efficiency (RE)} = \frac{\text{variance of estimator}}{\text{variance of estimator}}$$

Example:-

- Variance of estimator A is 10
- Variance of estimator B is 20

$$RE = \frac{\text{Variance A}}{\text{Variance B}}$$

$$= \frac{10}{20} = 0.5$$



## (ii)

Risk function:-

The risk function is a mathematical function that measures the expected loss or risk associated with a particular decision or action.

formula:-

$$R(\theta, \sigma) = E(L(\theta, \sigma))$$

Where:-

- $R(\theta, \sigma)$  is the risk function
- $\theta$  is the true parameter value
- $\sigma$  is the decision or action
- $L(\theta, \sigma)$  is the loss function
- $E()$  is the expected value operation

## (iii)

Drawbacks of factorization Theorem

The factorization theorem is a powerful tool in statistics, particularly in the context of exponential families and likelihood functions.

- Complexity
- Assumption
- limited scope
- Interpretation



The fertilization theorem remain  
a variable tool in the field  
of statistics.

### div

Exponential family of distribution.

The exponential family of distribution is a class of probability distribution that share a common mathematical form. These distribution are widely used in statistical that can be expressed in the form:-

$$f(x/\theta) = h(x) \cdot \exp[\eta(\theta)T(x) - A(\theta)]$$

Where:-

- $f(x/\theta)$  is a probability density
- $\theta$  is a parameter
- $\eta(\theta)$  is a natural parameter
- $T(x)$  is a sufficient statistics
- $h(x)$  is a base measure
- $A(\theta)$  is a log normalization



2018

Q13

Weak and strong Consistency?

• weak consistency:-

An estimator is weakly consistent if it gets closer to the true value as the sample size increase. Weak consistency means that the estimator is likely to be close to the true value.

• strong Consistency

An estimator is strongly consistent if it's almost certain to get closer to the true values as the sample size increase. The estimator is almost certain to converge to the true value.

Q14

Regularly Conditions

(1) linearity:- Relationship between variables is linear.

(2) Independence:- Each data point is separate and doesn't affect others.



Homoscedasticity: Equal variance across all levels of independent variables.

Normality:- Error term is normal distributed

5) No multicollinearity: Independent variables are not highly correlated

6) No autocorrelation: Error terms are not correlated with each other

7) full rank matrix: matrix of independent variables has full matrix.

8) Non-zero Variance: Error term has non-zero Variance

### 3b

#### Loss functions-

A loss function, also known as a cost function or objective function, is a mathematical function that measures the difference between predicted and actual values.

#### Purpose:-

The purpose of a loss function is to evaluate the performance of a machine learning model and guide the optimization process to minimize the error.



Q4)

Bayes Estimator:-

A Bayes estimator is a statistical estimator that minimizes the expected loss with respect to the posterior distribution of the parameter being estimated.

In simpler terms, it uses Bayes's theorem to update the probability estimates for a parameter as more evidence or information becomes available.

Step by step breakdown:-

- (1) prior distribution
- (2) likelihood function
- (3) posterior Distribution
- (4) loss function
- (5) Minimize Expected loss.



2019

i) Best Asymptotically Normal (BAN) estimators.  
The best normal estimator is an Asymptotically (BAN) estimator that has the smallest Asymptotically normal estimators.

Examples:-

Maximum likelihood estimator:-

The maximum likelihood estimator is often the (BAN) estimator for many parameter models.

Bayes Estimator:-

Under certain condition, the Bayes estimator can be the BAN estimator.

ii)

Property of Invariance:-

The property of invariance states that the estimator should remain unchanged under certain transformations or changes in the



## Types of invariance

- (1) Scale invariance
- (2) location invariance
- (3) Rotation invariance

Example:-

- (1) Mean estimator
- (2) median estimator
- (3) Standard deviation estimator

iii)

Risk function  $\rightarrow$  repeat

iv)

Efficient Estimator :-

An efficient estimator is a statistical estimator that achieves the lowest possible variance among all unbiased estimators of a parameter.

Key points

- (1) Unbiasedness
- (2) Minimum Variance
- (3) Cramér-Rao lower bound



Example:-

The Sample mean is an efficient estimator of the population mean in a normal distribution

2020

Repeat  $\rightarrow$  2021  
Repeat  $\rightarrow$  2022

2021

What is the problem of estimator? How it can be dealt?

(1) Bias:- Estimator's expected value doesn't equal to the true parameter

Solution Use unbiased estimator  
or apply bias correction

(2) Variance High variability in estimators

Solution:- Use efficient estimators  
or increase sample size

(3) Consistency Estimator doesn't converge to the true value with large sample size

Solution Ensure condition for

Consistency are met. Estimator is



Solution Use robust estimators

like the median

(5) Efficiency:- Estimator has higher Variance than the minimum possible

Solution Use efficient estimators that achieve the Cramer-Rao lower bound.

Refact 201

(ii)

→ Difference between:-

(1) Asymptotic unbiasedness and Consistency

(2) Efficiency and Mean Square Consistency

Solution:-

(1) Asymptotic Unbiasedness Consistency

An estimator is asymptotically unbiased if its expected value converges to the true parameter value as the sample size increases

An estimator is consistent if it converges in probability to the true parameter value as the sample size increases



## (2) Efficiency

An estimator is efficient if it has the smallest variance among all unbiased estimator

Mean Square Consistency  
An estimator is mean square consistency if its mean square error (MSE) Converges to zero as the sample size increase.

## (iii)

Compare the best asymptotically Normal Estimator, Minimum Variance Bound estimator and efficient estimator?

Answer:-

- Asymptotically Normal Estimator:- As the sample size get really big, this estimator's distribution look like a normal dis.
- Minimum Variance Bound Estimator:- This is the best unbiased estimator because it has the smallest possible variance among all unbiased estimators.
- Efficient Estimator:- This is an estimator that achieves the smallest possible variance, making it the most precise. it's basically an MVUE that hits the lowest variance limits.



QVI

state Lehmann - Scheffe's theorms.

The Lehmann - Scheffe'

Theorem helps us find the best way to estimate something. if we have a special kind of data summary and an unbiased guess, we can use them to get the most accurate and reliable estimate possible.

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Lehmann - Scheffe



2021

Write a note on the Comparison of Asymptotic Unbiasedness and Consistency?

Defination  $\rightarrow$  repeat

focus:- reducing bias over time

focus:- getting more accurate over time

Example:-

Example:-

Imagine you're guessing the average height of people in a city. As you ask more and more people, your guessing gets closer to the actual average height.

Imagine you're guessing the average height of people in a city. As you ask more and more people, your guess becomes more accurate and reliable.

ii)

Write the procedures for finding whether any estimator is sufficient or not, what are (if any) the objections on them?

**Procedure:-**

(1) choose a statistic:- Pick a Summary of your data (statistics)



## (2) Factorization Theorem:-

check if you can  
Split the probability function into two parts

- One part depend on statistics and the parameter
- The other part depends only on the data.

## Objections:-

### (1) Complexity:-

The process can be complicated

### (2) Interpretation:-

it can be hard to understand

### (3) practically:-

Sometimes it's not easy to find a sufficient statistic in real-world situations.

can repeat

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How would you explain the joint Sufficiency?

Joint Sufficiency refers to a situation where a set of statistics together provide all the information about a parameter.



Example:-

Imagine you have a dataset. you calculate two statistics from it say  $T_1$  and  $T_2$ . if knowing both  $T_1$  and  $T_2$  gives you all the information about the parameter  $\theta$ , the  $T_1$  and  $T_2$  are joint sufficient for  $\theta$ .

repeated-2022

2021 repeat

div

Differentiate between location invariant and scale invariant estimators?

location invariant estimators

Scale invariant estimator

These estimators are not affected by changes in the location (or shift) of the data. if you add a constant to all data point, the estimator remain the same.

These estimators are not affected by change in the scale of the data. if you multiply all data points by a constant, the estimator remains the same.

Example:- The Sample

Variance is location invariant because adding a constant to all data point does not change

Example:- The Sample correlation

Coefficient is scale invariant because multiplying all data point by a constant does not change correlation.



Repeat → 25/11

Q17

What do you mean by Completeness?  
Explain.

In statistics Completeness refers to a property of a statistic that ensures it captures all the information about a parameter.

Meaning:- A statistic is complete if it captures all the information about a parameter.  
Why it matters:- it ensures you have all the information needed for accurate estimation.

Q18

Repeat → 26/11

write and explain different forms of efficiency.

## Relative Efficiency:-

- Meaning:- Compares the variance of two unbiased estimators
- Example:-

Estimator A is more efficient than B if it has a smaller variance.



(V)

What do you mean by Completeness?  
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(VI)

Recall 2021

write and explain different forms of efficiency:-

## Relative Efficiency:-

- Meaning:- Compares the variances of two unbiased estimators
- Example:-

Estimator A is more efficient than B if it has a smaller variance



## Asymptotic Efficiency:-

- Meaning:- Efficiency of an estimator as the sample size approaches infinity
- Example:- Maximum likelihood Estimators (MLEs) are often asymptotically efficient.

## Finite Sample

- Meaning:- Efficiency of an estimator for a fixed, finite sample size
- Example:- Some estimators performs better with small sample size.

~ (2021) ~

Paper No 2.

all the S/Q repeat.

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(2022)

Q11B

Repeat

Q11B

What is the importance of sufficiency and joint sufficiency as a property of good estimators?

### Importance of Sufficiency:-

Complete information:- A Sufficient statistic captures all the information about a parameter.

Efficiency:- Leads to more reliable and precise estimators.

Simplification:- Reduced data to a single statistic without losing information.

### Importance of joint Sufficiency:-

Combined information:- multiple statistics together provide all the information about a parameter.



improved Estimation:- Lead to better and more accurate estimates and flexibility:- Uses multiple statistics to capture different aspects of the data.

iii)

The Cramer-Rao inequality is basis of efficiency, discuss briefly?

Cramer Rao lower bound:-

it states that the variance of any unbiased estimator is at least as large as the inverse of the fisher information.

In mathematically, if  $\hat{\theta}$  is an unbiased estimator of a parameter then:-

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

where  $I(\theta)$  is the fisher information



div  
Compare best asymptotically normal estimator with efficient estimator?

## Best Asymptotically Normal Estimator:-

- Meaning:- A sample size gets really big, the estimator's distribution looks like a normal distribution.
- Focus:- it becomes approximate normal which simplifies inference.

## Efficient Estimator:-

- Meaning:- The estimator achieves the smallest possible variance among all unbiased estimators.
- Focus:- it reaches the Cramer-Rao lower bound, making it the most precise and reliable.



Q(VI)  
How the property of unbiasedness and efficiency are related? Which one is more needed?

## Relationship between Unbiasedness and efficiency

Unbiasedness:-

An estimator is correct on average.

Efficiency:- An estimator has the smallest possible variance, making it the most precise.

Which is more needed?

Content - Dependent:-

Both are important, but efficiency is often more desirable because it leads to more precise estimates.



and divided  
by the number  
of people

(2023)

(Q.1)

Define Estimator, Estimate and estimation with example

Estimator:- An estimator is a statistical formula or rule used to calculate an estimate of a population parameter based on a sample data.

Example:- To guess the average height of all people in a country, we use a formula that adds up the heights of a small group of people.

Estimate:- An estimate is a specific value calculated from a sample of data using an estimator.

Example:- Using the formula we calculate the average height to be 175.2 cm. This is our estimate.

Estimation:- Estimation is the process of using an estimator to calculate an estimate of a population parameter based on a sample of data.

Example:- We calculate data on height, use the formula, and get an answer. This is

estimate → Estimator → formula  
Estimate → Answer Estimation → Process of getting the answer



(b)

Describe the type of consistency?  
Repeat 2018 (i)

(c)

How Risk and loss function are related?

loss function repeat 2018 (3)

Risk function repeat 2017 (11b)

(d)

Define exponential family distribution?  
Repeat  $\rightarrow$  2017 (iv)

(e)

Describe the role of mean-square error in estimation theory.

(MSE)  $\rightarrow$  repeat 2020 (iv)

estimation theory. repeat 2019 (iv)

(f)

What is meant by sufficiency? Explain the Fisher-Neyman factorization criterion for sufficiency?



What is meant by sufficient?

repeat  $\rightarrow$  2021 (III)

Fisher-Neyman factorization -

Definition:-

The Fisher-Neyman factorization is a mathematical technique used in statistics to factorize the likelihood function of a statistical model.

The Fisher-Neyman factorization involves factoring likelihood function  $L(\theta; x)$  into two components

$$L(\theta; x) = g(T(x); \theta) h(x)$$

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